

JESUS PEREZ BAZAROT

Seville, Spain | jp.bazarot@gmail.com

linkedin.com/in/jpbazarot | github.com/PersusUS | persus.netlify.app

SUMMARY

Fourth-year Computer Engineering student at the University of Seville. **Expected graduation: 06/2028.** Wrote five CUDA implementations of the selective state space recurrence, ending in a single-pass decoupled look-back scan that sustains **79.6–91.5% of theoretical memory bandwidth** and runs **54x–69x** faster than the best pure-PyTorch baseline. Reached the same recurrence from the framework side first: diagnosed Mamba’s CUDA kernels as untraceable under `torch.compile` fullgraph and replaced the sequential $O(L)$ recurrence with a parallel associative scan, **48x measured speedup** (OpenAI Parameter Golf). Sole author of a continual-learning benchmark paper under review at a NeurIPS 2026 workshop, with 375 experimental runs, code and all result files public. Seeking a Summer 2027 deep learning software internship.

PROJECTS

Selective-scan CUDA kernels | *CUDA C++*, github.com/PersusUS/selective-scan-cuda 2026

- Implemented the selective state space recurrence $h_t = a_t h_{t-1} + b_t$ as five CUDA kernels, from a shared-memory Brelloch scan up to a single-pass decoupled look-back scan (Merrill and Garland, NVIDIA NVR-2016-002), the algorithm behind CUB’s `DeviceScan`
- Fastest kernel sustains **79.6–91.5% of theoretical DRAM bandwidth** on an RTX 4050 Laptop (192 GB/s): **54x–69x** over the best pure-PyTorch baseline and 2.18x–2.48x over the three-pass version, against 2.33x predicted by a memory-traffic model written before the kernel existed
- Built the inter-block protocol on `atomicAdd`, `__threadfence` and volatile flag loads, claiming tiles by atomic counter rather than `blockIdx` so no tile waits on an unscheduled predecessor; a warp-parallel look-back over shuffle and ballot intrinsics cut chain latency by up to 1.80x at 256 tiles
- Verified with **107 correctness checks** against a float64 reference and **630 determinism runs** to expose intermittent races; of five performance predictions registered before measuring, three held and two were refuted (a pre-allocated workspace changed nothing), both documented as refuted in the repository

HybridMamba-11, OpenAI Parameter Golf | *PyTorch, CUDA, torch.compile, openai/parameter-golf #1365* 2026

- Designed HybridMamba-11, a hybrid Mamba state space model and Transformer architecture (11 layers) with U-Net skip connections, the first state space model submission in the competition
- Diagnosed the core bottleneck: Mamba CUDA kernels are untraceable under `torch.compile` fullgraph mode: roughly 300 training steps in the 600s budget against about 20,000 for top entries
- Replaced the sequential $O(L)$ recurrence with a parallel associative scan (Hillis-Steele) in pure PyTorch: $O(\log L)$ depth, fullgraph compilable, **48x measured speedup** on the optimized path
- Packaged the submission at 31.8M parameters and 13.6MB compressed against a 16MB cap, after removing 12.5M dead parameters and fixing compile, dtype and state-dict compatibility issues; entry open and preliminary, pending H100 time to validate the fullgraph path

Perseo, personal AI assistant | *Tauri 2, Rust, React, Python, Ollama, Gemini Live API* 2023 – Present

- Multi-agent assistant running as a Windows service: local-model triage (Qwen3 4B on Ollama), real-time dual-stream voice over a Gemini Live WebSocket with barge-in and wake-word detection, and a Rust backend with no cloud persistence; hardened to 919 unit tests, 17 end-to-end verifiers and green CI

PUBLICATIONS

Perez Bazarot, J. **Forgetting in World Models Does Not Follow Task Distance.** Submitted to the CL4FMAgents workshop at NeurIPS 2026 (8 pages, non-archival, under review). Full 26-page version, code, and all result files: github.com/PersusUS/WorldModelsBenchmark

- Built a component-level continual-learning benchmark isolating the transition component of a world model: 375 runs over 3 environment families (MiniGrid, MuJoCo/Gymnasium, DeepMind Control) by 3 dynamics-distance levels by 5 methods (finetuning, EWC, infinite replay, progressive networks, and UG-MTM, a proposed uncertainty-gated mixture of transition models), up to 10 seeds per cell; same-seed reruns reproduce every metric bit-for-bit
- Result 1: the labelled task-distance axis carries no rank information about forgetting (Spearman rho = 0.00). Forgetting peaks at the intermediate level in all three families, and the effect survives a rerun at twice the training budget

- Result 2: forgetting concentrates in the encoder, where isolated metrics do not look. Finetuning degrades task-A reconstruction by 811x while the transition metric reports an apparent improvement, and EWC's Fisher information is exactly zero over the encoder by construction
- Both headline results are negative, including one that refutes my own proposed architecture; next archival target is CoLLAs 2027

EXPERIENCE

OrgaAI Jun 2025 – Sep 2025
AI/ML Developer *Spain*

- Owned the audio processing and transcription pipeline of a confidential conversational AI product
- Cut pipeline runtime by 10x by reusing existing dataset transcripts instead of running full WhisperX passes
- Researched and prototyped a Slot Attention integration for model architecture improvements (under NDA)

Minerva Research Group, University of Seville 2023 – 2024
Research Intern *Seville, Spain*

- Selected as an internal student researcher in first year, an early-entry position normally offered to later-year students
- Built machine learning classifiers over gas-emission datasets from Iberian ham samples to separate feeding categories (cebo vs. bellota)

ByTheWay Carpool (startup) 2024 – 2025
Co-Founder and Co-CTO *Spain*

- Co-founded a carpooling platform for Loyola University students and co-led its technical architecture: rider-driver matching, trip coordination and operational tooling

NetKey (technology startup) 2023 – 2024
Chief Technology Officer *Seville, Spain*

- Ran the full technical stack as sole engineer: automated order handling, client communications, internal email workflows and the company website end to end, including CI/CD

EDUCATION

University of Seville 2023 – **06/2028** (expected)
B.Sc. in Computer Engineering, Computer Engineering specialization *Seville, Spain*

- Selected coursework: Linear and Numerical Algebra, **Matricula de Honor** (top Spanish grade, capped by national regulation at 5% of a course); Discrete Mathematics, 9.5/10

Beijing Institute of Technology (BIT) Sep 2025 – May 2026
Erasmus exchange, Computer Science and Big Data Analysis *Beijing, China*

- Coursework: Machine Learning, Big Data Analysis, Digital Logic

SKILLS

GPU and systems: CUDA C++ (shared memory, warp shuffles, atomics, memory fences, occupancy and bandwidth analysis), parallel scan algorithms, PyTorch C++ extensions, Triton, `torch.compile`, C, C++, mixed precision (BF16), quantization, distributed data parallel training, Linux, Docker, Git, CI/CD

Machine learning: PyTorch, Transformers, state space models (Mamba), world models (RSSM), continual learning (EWC, replay, progressive networks), retrieval-augmented generation

Research and evaluation: performance modelling, benchmark design, ablation studies, seed and reproducibility protocols, LLM evaluation, pandas, NumPy, matplotlib

Languages: Spanish (native), English (C1, Trinity 2020), Chinese (conversational)

AWARDS AND ACTIVITIES

Wanted Entrepreneurship Competition — first place out of 20 teams 2023
Algorithmics Club, University of Seville — member 2023 – Present